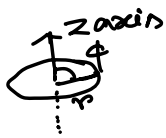
① Eigenfunction $\psi = \psi(\phi)$

(i) 

r = fixed $\phi = 0$ to 2π

As 'r' is constant for a circle of radius 'r'

$\therefore$ Schrodinger equation

K.E. operator $= \dfrac{\hat{L}_z^2}{2I}$ $\qquad I$ = Moment of Inertia $= mr^2$

$\hat{L}_z = -i\hbar \dfrac{\partial}{\partial \phi}$

$$\therefore \boxed{-\frac{\hbar^2}{2I}\frac{\partial^2 \psi}{\partial \phi^2} = E\psi}$$ (Time independent Schrodinger equation)

(ii) $$-\frac{\hbar^2}{2m}\frac{\partial^2 \psi}{\partial \phi^2} = E\psi$$

As $\hat{L}_z \psi = m\hbar\psi$

where $\psi = \psi(\phi) = \dfrac{1}{\sqrt{2\pi}} e^{im\phi}$

$m = 0, \pm 1, \pm 2 \ldots$

So, $\dfrac{\hat{L}_z}{2I}\psi(\phi) = m^2\hbar^2\psi(\phi)$

$m = 0, \pm 1, \pm 2 \ldots$ (allowed values)

The allowed eigenvalues are $m^2\hbar^2$

eigenfunctions $\dfrac{1}{\sqrt{2\pi}} e^{im\phi}$

(2) For diatomic molecule with fixed bond length 'r'

$\therefore$ The laplacian

$$\nabla^2 = \left[\frac{1}{r^2}\frac{\partial}{\partial r}\left(r^2\frac{\partial}{\partial r}\right) + \frac{1}{r^2 \sin\theta}\frac{\partial}{\partial \theta}\left(\sin\theta\frac{\partial}{\partial \theta}\right) + \frac{1}{r^2 \sin^2\theta}\frac{\partial^2}{\partial \phi^2}\right]$$

$= 0$

for fixed bond length r

$$\therefore \; -\frac{\hbar^2}{2m}\frac{1}{r^2}\underbrace{\left[\frac{1}{\sin\theta}\frac{\partial}{\partial\theta}\left(\sin\theta\frac{\partial}{\partial\theta}\right)+\frac{1}{\sin^2\theta}\frac{\partial^2}{\partial\phi^2}\right]}_{-\hat{L}^2/\hbar^2}\psi(\theta,\phi) = E\psi(\theta,\phi)$$

$$\frac{\hat{L}^2\psi(\theta,\phi)}{2I} = E\psi(\theta,\phi)$$

$\psi(\theta,\phi) = Y_l^m(\theta,\phi) =$ eigenfunctions for $\hat{L}^2$ operator.

So, $$\frac{\hat{L}^2 Y_l^m(\theta,\phi)}{2I} = \frac{l(l+1)\hbar^2}{2I}Y_l^m(\theta,\phi)$$

eigenvalues are $\frac{l(l+1)\hbar^2}{2I}$ $l=0, 1, 2, 3 \cdots$

eigenfunctions are $Y_l^m(\theta,\phi)$ where $m=-l, -l+1 \cdots 0, 1, \cdots +l$

(iii) The consecutive energy difference

$$\Delta E = \frac{(l+1)(l+2)\hbar^2}{2I} - \frac{l(l+1)\hbar^2}{2I}$$

$$= \frac{\hbar^2}{2I}(l+1)\left[l+2-l\right]$$

$$= \frac{\hbar^2(l+1)}{I} \qquad l=0, 1, 2, \cdots$$

4. $$\int_0^{2\pi}\int_0^{\pi} Y_1^1(\theta,\phi)\, Y_2^0(\theta,\phi)\, \sin\theta \, d\theta \, d\phi$$

Do the integration and show the value will come to zero.

(5) $[L_x, L_z] = i\hbar L_y$ (You can carry out the commutation and show the relations

$$\vec{L} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ x & y & z \\ p_x & p_y & p_z \end{vmatrix}$$

$$[\hat{L}^2, L_y] = 0$$

$$[\hat{L}^2, L_y + L_x + L_z] = 0$$

(6) $\hat{L}^2 Y_3^2(\theta, \phi) = 3(3+1)\hbar^2 Y_3^2(\theta, \phi)$

$l = 3$, $m = 2$ $\quad L_z Y_3^{-2}(\theta, \phi) = -2\hbar Y_3^{-2}(\theta, \phi)$

(7) $\Delta E = \dfrac{\hbar^2 (l+1)}{I}$

$l = 0$ for $l = 0$ to $l = 1$ transition

$= \dfrac{\hbar^2}{\mu r^2}$

$$\mu = \left(\frac{m_1 m_2}{m_1 + m_2}\right) \times 1.66 \times 10^{-27} \text{ kg}$$

$$r = 2 \times 10^{10} \text{ meter}$$

$$\Delta E = h\nu$$

$$\boxed{\nu = \frac{\hbar^2}{h \mu r^2}} \text{ per sec}$$

(8) (i) (ii) (iii) you can find in book.

(iv) Average K.E. for g.s. of H.O.

$$I_1 = \int_{-\infty}^{\infty} \psi_0 \, \frac{-\hbar^2}{2m} \frac{\partial^2}{\partial x^2} \psi_0 \, dx$$

$$\psi_0 = \left(\frac{\alpha}{\pi a_0^3}\right)^{1/2} e^{-\alpha x^2/2}$$

Averge P.E.

$$I_2 = \int_{-\infty}^{\infty} \psi_0 \, \tfrac{1}{2} m \omega^2 x^2 \psi_0 \, dx$$

Show $I_1 = I_2$

(v)

$$\Delta x = \left[\langle x^2 \rangle - \langle x \rangle^2 \right]^{1/2}$$

$$\Delta p_x = \left[\langle \hat{p}_x^2 \rangle - \langle \hat{p}_x \rangle^2 \right]^{1/2}$$

Evaluate the expectation values of these operators and find $\Delta x \, \Delta p_x \simeq \hbar$

(9)

(i) $$-\frac{\hbar^2}{2m_e} \nabla^2 \psi(r, \theta, \phi) - \frac{e^2}{4\pi\epsilon_0 r} \psi(r, \theta, \phi) = E \psi(r, \theta, \phi)$$

(ii) $[\hat{H}, \hat{L}^2] = 0$ done in class

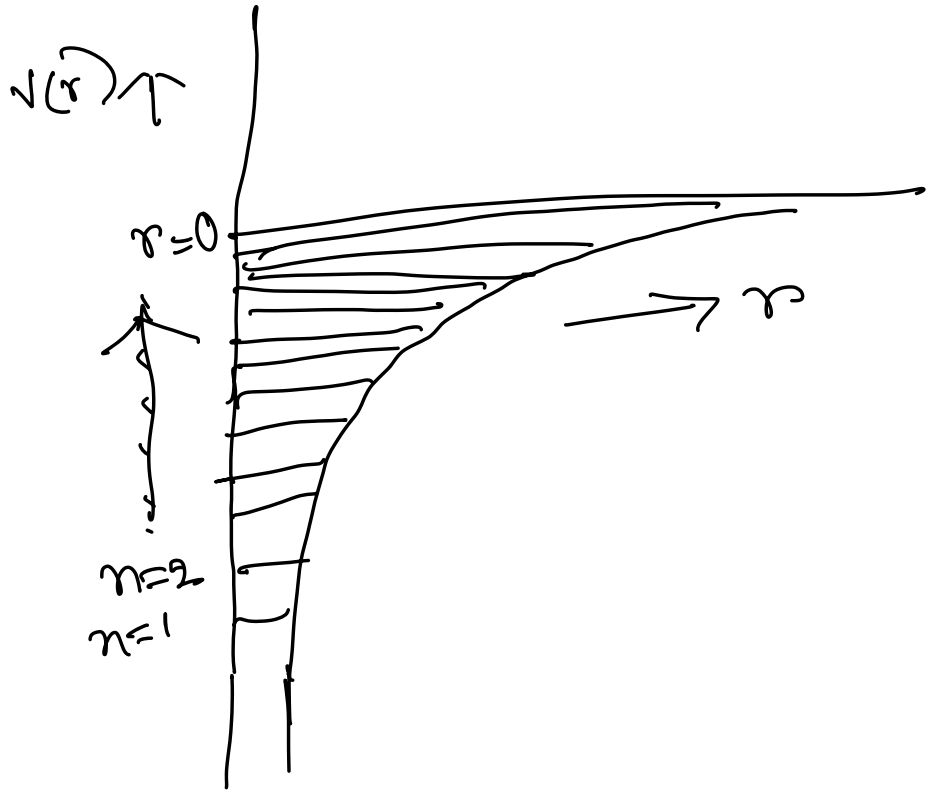
(iv) $$\boxed{E_n = -\frac{Z^2 \, 13.6 \text{ eV}}{n^2}} \quad n = 1, 2, 3 \ldots$$

(iii) For hydrogen $Z = 1$

$$E_H = -\frac{13.6}{n^2} \text{ eV}$$

$n = 1, 2, 3 \ldots$

(iv) $$V(r) = \frac{-e^2}{4\pi\epsilon_0 r}$$



(v) for $n = 2$

Degree of degeneracy $= 2$
$= 4$

(vi) $\psi_{nlm}(r,\theta,\phi) = R_{nl}(r)\, Y_l^m(\theta,\phi)$

$n=1$ $\quad l=0,\ m=0$ (g.s.)

$n=2,$ $\quad l=0,\ m=0$

$\quad\quad\quad\quad l=1,\ m=0,\ \pm 1$

Degeneracy $2^2=4$

$n=3$ $\quad l=0,\ m=0$

$\quad\quad\quad\quad l=1,\ m=0,\ m=\pm 1$

Degeneracy $\quad l=2,\ m=0,\ m=\pm 1,\ m=\pm 2$

$3^2=9$ and so on

(10)

$$\langle r\rangle = \int_0^{\pi}\int_0^{\infty}\int_0^{2\pi} \psi_{000}^{*}\, r\, \psi_{000}\, r^2 \sin\theta\, d\theta\, dr\, d\phi$$

$$= 4\pi \int_0^{\infty} \psi_{000}^{*}\, r\, \psi_{000}\, dr$$

$$= 3/2\, a_0 \quad \text{(after integration)}$$

as $\int_0^{\pi} \int_0^{2\pi} \sin\theta \, d\theta \, d\phi = 4\pi$